

Supplementary Methods

This document provides implementation details supporting reproducibility of BERTRAG.

Corpus construction. Documents were collected from Awori and Ogu archival sources, oral histories, community narratives, ritual descriptions, proverbs, and community histories. Materials were transcribed, translated where necessary, segmented into semantically coherent passages, and annotated with structured metadata.

Domain adaptation. A pretrained BERT encoder was adapted using masked language modelling and optional contrastive fine-tuning on the indigenous knowledge corpus.

Retrieval and generation. Hybrid semantic retrieval combines dense embeddings with metadata-aware filtering. Retrieved evidence is incorporated into the LLM prompt with provenance constraints. The system abstains when evidence is insufficient or culturally restricted.

Evaluation. Retrieval metrics (Recall@10, MRR, nDCG@10), faithfulness, hallucination, attribution precision, abstention behaviour, and expert cultural fidelity were evaluated using 260 held-out queries.

Statistics. Paired bootstrap testing (10,000 resamples), 95% bootstrap confidence intervals, exact p-values, and Cohen's d effect sizes were computed for the principal comparisons.